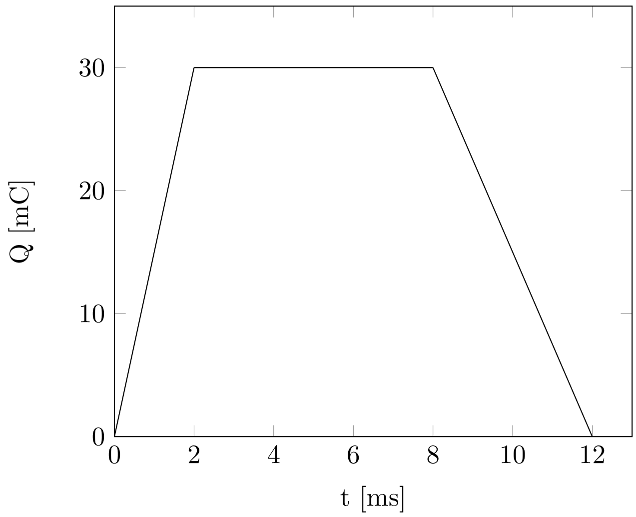
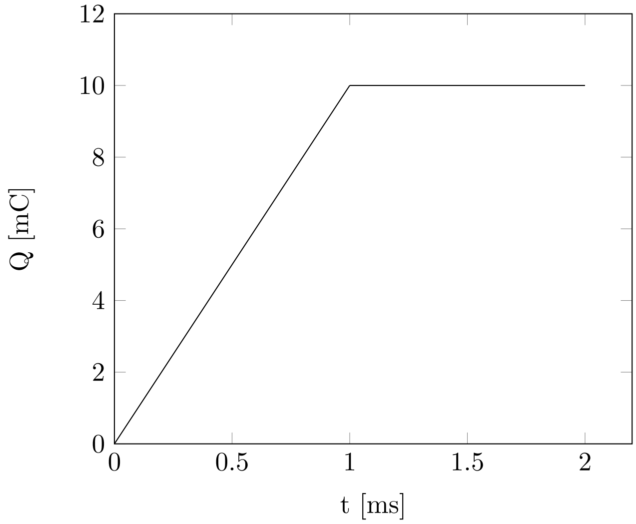
Tutorial 1

Question 1.4
A current of 7.4 A flows through a conductor. Calculate how much charge passes through any cross-sectional area of the conductor in 20 s.
Memo: $Q = \int_{t_1}^{t_2} I(t)dt$ $Q = \int_{0}^{20} 7.4dt$ $Q = [7.4t + c]_0^{20}$ $Q = 148C$

Question 1.6
The charge entering a certain element is shown below. Find: a) $t = 1\ ms$ b) $t = 6\ ms$ c) $t = 10\ ms$ 
a) $$I = \frac{dQ}{dt} = \frac{30-0}{2-0} = 15A$$
b) $$I = \frac{dQ}{dt} = 0A$$
c) $$I = \frac{dQ}{dt} = \frac{0-30}{12-8} = -7.5A$$

Question 1.8

The current flowing past a device is shown below. Calculate the total charge flowing past the point.



Memo: Domain 0:1

$$Q = \int_0^1 10t \; dt$$

$$Q = [5t^2 + c]_0^1 = 5 \times 10^{-3} \times (1 \times 10^{-3})^2 = 5 \; \mu C$$

Domain 1:2

$$Q = \int_1^2 10 \; dt$$

$$Q = [10t + c]_1^2 = 10 \times 10^{-3} \times 1 \times 10^{-3} = 10 \; \mu C$$

Total: $Q = 15 \; \mu C$

Question 1.11
A rechargeable flashlight is capable of delivering 90 mA for a period of about 12 hours. How much charge can it release at this rate? If its terminal voltage is 1.5 V, how much energy can the battery deliver?
Memo: Charge: $$Q = \int_0^{12\times 60\times 60} 0.09\ dt$$ $$Q = 0.09 \times 12 \times 60 \times 60 = 3.888\ kC$$ Energy: $$W = \int_{Q_1}^{Q_2} V\ dQ$$ $$W = \int_0^{3888} 1.5\ dQ$$ $$W = 1.5 \times 3888 = 5.832\ kJ$$

Question 1.12

The current flowing through an element is given by

$$i(t) = \begin{cases} 3t\ A & 0 \leq t < 6\ s \\ 18\ A & 6 \leq t < 10\ s \\ -12\ A & 10 \leq t < 15\ s \\ 0 & t \geq 15\ s \end{cases}$$

Plot the charge stored in the element over $0 \leq t < 20\ s$.

$$q(t) = \int_{t_1}^{t_2} I(t)\ dt$$

For $0 \leq t < 6\ s$

$$q(t) = \int_{0}^{6} 3t\ dt$$

$$q(t) = 1.5t^2 + C_1$$

$$q(0) = 0$$

$$C_1 = 0$$

For $6 \leq t < 10\ s$

$$q(t) = \int_{6}^{10} 18\ dt$$

$$q(t) = 18t + C_2$$

$$q(6) = 1.5 \times 6^2 = 54$$

At t = 6, q(t) = 54

$$\therefore C_2 = -54$$

For $10 \leq t < 15\ s$

$$q(t) = \int_{10}^{15} -12\ dt$$

$$q(t) = -12t + C_3$$

$$q(10) = 18 \times 10 - 54 = 126$$

At t = 10, q(t) = 126

$$\therefore C_3 = 246$$

For $t \geq 15\ s$

$$q(t) = \int_{15}^{20} 0\ dt$$

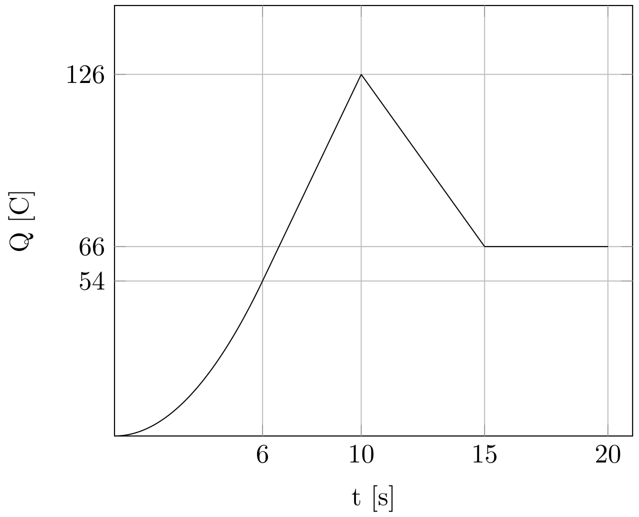
$$q(t) = C_4$$

$$q(6) = -12 \times 15 + 246 = 66$$

At t = 15, q(t) = 66

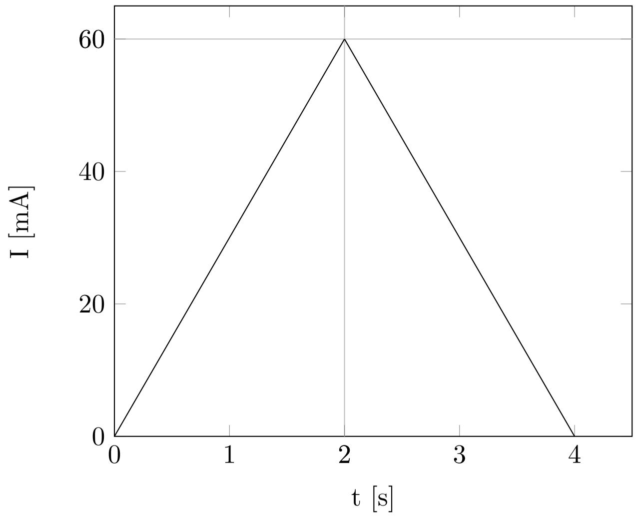
$$\therefore C_4 = 66$$

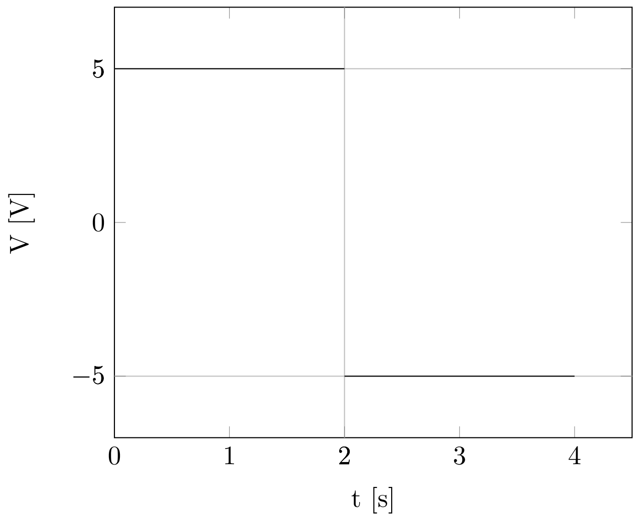
Thus the graphs is:



Question 1.16

The following graphs show the voltage across and current through an element:





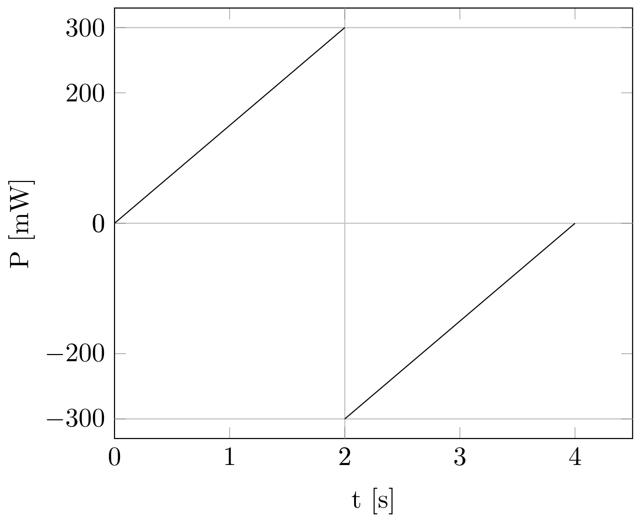
a) Sketch the power delivered to the elements for $t > 0\ s$.

b) Find the total energy absorbed by the element for a period of $0 \leq t < 4\ s$.

a)
For $0 \leq t < 2$

V = 5V
I = 30t mA
$P = VI$
$P = 5 * 0.03t = 150t\ mW$

For $2s \leq t < 4s$
V = -5V
I = -30t +120 mA
$P = VI$
$P = -5 * (-0.03t + 0.12) = 150t - 600\ mW$



b)

$$W(t) = \int_{t_1}^{t_2} P(t)\ dt$$

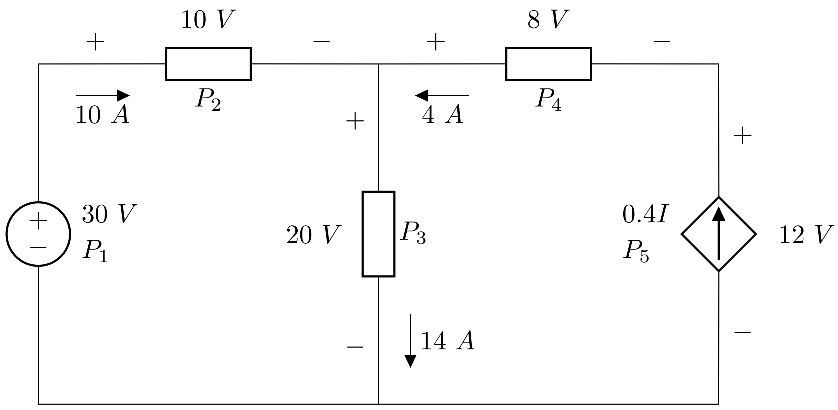
$$W(t) = \int_0^2 0.15t\ dt + \int_2^4 -0.15t + 0.6\ dt$$

$$W(t) = [0.075t^2]_0^2 + [-0.075t^2 + 0.6t]_2^4$$

$$W(t) = [0.075t^2]_0^2 + [-0.075t^2 + 0.6t]_2^4$$

$$W(t) = 0.3 - 1.2 + 0.9 = 0\ J$$

Question 1.18
Find the power absorbed by each of the elements.



Memo:

$$P_1 = -VI = -(30)(10) = -300\ W$$

$$P_2 = VI = (10)(10) = 100\ W$$

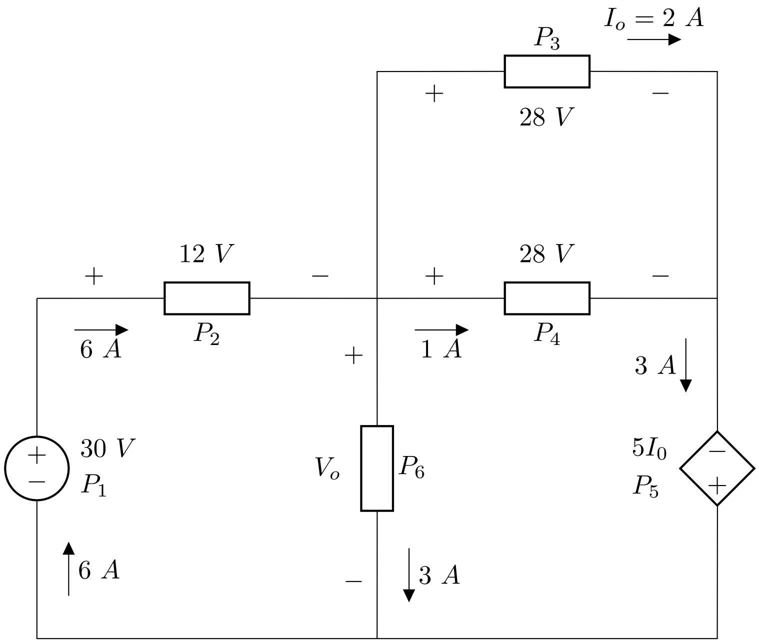
$$P_3 = VI = (20)(14) = 280\ W$$

$$P_4 = -VI = -(8)(4) = -32\ W$$

$$P_5 = -VI = -(12)(4) = -48\ W$$

Question 1.20

Find V_o and the power absorbed by each of the elements.



Memo:

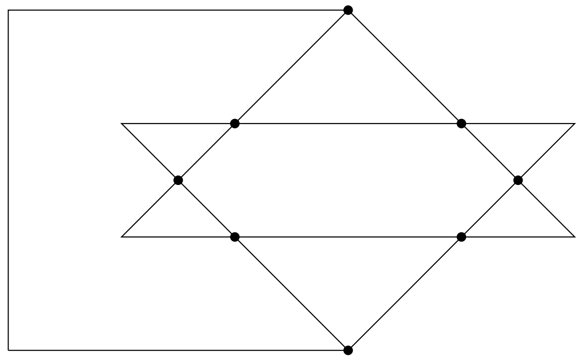
$$P_1 = -VI = -(30)(6) = -180\ W$$
$$P_2 = VI = (12)(6) = 72\ W$$
$$P_3 = VI = (28)(2) = 56\ W$$
$$P_4 = VI = (28)(1) = 28\ W$$
$$P_5 = -VI = -(5 \times 2)(3) = -30\ W$$
$$\sum P = 0$$
$$-180 + 72 + 56 + 28 - 30 + P_6 = 0$$
$$P_6 = 54 = 3V_o$$
$$\therefore V_o = 18V$$

Question 1.24
A utility charges 8.2 cents/kWh. If a consumer operates a 60 W light bulb continuously for one day, how much is the consumer charged?
Memo: $$W \ in \ kW.h = kW \times h = \frac{60}{1000} \times 24 = 1.44 \ kW.h$$ $$Cost = 1.44 \times 8.2 = 11.808 \ cents$$

Question 1.28
A 60 W incandescent lamp is connected to a 120 V source and is left burning continuously in an otherwise dark staircase. Determine: a) the current through the lamp b) the cost of operating the light for one non-leap year if electricity costs 9.5 cents per kWh.
Memo: a) $$I = P/V = 60/120 = 0.5\ A$$ b) $$W = Pt = \frac{60}{1000} \times (365)(24) = 525.6\ kW.h$$ $$cost = 525.6 \times 9.5$$ $$cost = R49.93$$

Question 2.6

In the network graph below, determine the number of branches and nodes.



Memo:

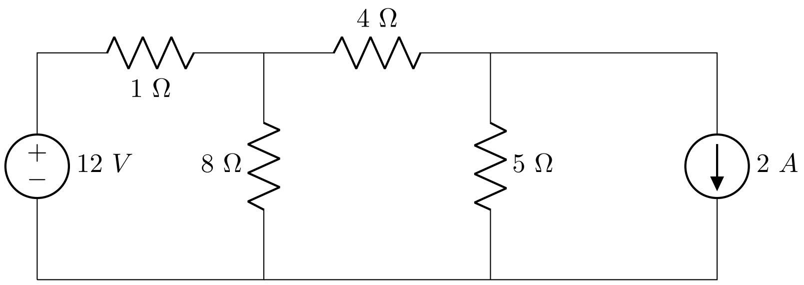
$$Nodes = 8$$

$$Loops = 8$$

$$Branches = n + l - 1 = 15$$

Question 2.7

Determine the number of branches and nodes in the circuit



Memo:

$$Branches = 6$$

$$Nodes = 4$$